Finite-size decay-mode hybridization near a mean-field fold in an open Rydberg ensemble

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Mean-field multistability and the spectrum of a finite open quantum system describe different notions of degeneracy. We investigate their relation in a uniformly interacting Rydberg ensemble with independent local decay, retaining the full permutation-invariant operator space. On the monostable side of a mean-field fold, two real decay eigenvalues exhibit an avoided crossing with a rapidly decreasing finite-size gap. Physical product-state perturbations show that the high-branch relaxation response transfers between these modes across the crossing, while an oscillatory pair can decay more slowly than either real mode. A biorthogonal reduction identifies the normalization-invariant product of cross couplings that determines the gap; population tails alone do not determine its scaling. Eliminating nonzero coherence orders yields an exact spectral-parameter-dependent operator. The two decay eigenvalues are recovered as self-consistent roots of one effective eigenvalue branch, with full-generator residuals below $1.2\times10^{-13}\gamma$ for $N = 24$–48. A local quadratic description reproduces the half-gap to relative errors decreasing from 1.2% to 6.6×10^{-6}. We also verify exceptional points under complex-detuning continuation. These results establish a finite-size spectral description and its dynamical limits without identifying the observed gap exponent with a quasipotential or asserting an asymptotic fold correspondence.

I. INTRODUCTION

Driven Rydberg ensembles connect nonlinear mean-field bistability with finite quantum dynamics [1–4]. A stable mean-field branch terminates at a saddle-node fold, whereas a finite quantum system evolves under a linear, trace-preserving Liouvillian. Relating these descriptions requires distinguishing stationary mean-field roots, eigenvalues of their stability Jacobian, and eigenvalues of the many-body generator.

This distinction matters for exceptional-point (EP) interpretations of nonlinear switching and stationary-root singularities [4–6]. A fold of stationary solutions need not be a defective eigenvalue of the mean-field Jacobian. The stationary sector of a finite-dimensional quantum dynamical semigroup cannot contain a nontrivial Jordan block, which would produce unbounded polynomial growth [7]. In the spectral-separation framework, metastability involves low-lying decay modes separated from faster relaxation [8, 9]. More generally, large eigenmode-expansion coefficients can prolong transients, so decay eigenvalues alone do not determine relaxation times [10, 11].

For uniformly interacting spins with local decay, collective switching and steady-state correlations have been studied [1, 2], and an exact finite-N steady state is available [12]. Exponentially slow switching is central to first-order dissipative transitions [13]; quantum-instanton methods predict switching rates for collective spin systems, including local dissipation [14, 15]. A related conjecture connects a symmetry-selected steady-state purification potential to the ordinary Liouvillian gap [16]. These results concern relaxation toward the stationary sector, rather than the separation of two modes with a finite common decay rate.

Finite-size avoided crossings from mode mixing [17] and few-mode descriptions of hysteresis [18] also have precedents. Cabot, Giorgi, and Zambrini connected a Liouvillian EP to a mean-field bifurcation between phase diffusion and quantum activation in a driven oscillator [19]. Collective-spin models exhibit spectral doublets approaching finite nonzero decay rates [20], with preparation-dependent visibility and cancelling spectral residues [21]. For the locally damped Rydberg model, we ask how a real-mode avoided crossing redistributes physical response and is represented after eliminating coherences.

We address this question in the complete permutation-invariant operator space at $\Omega = 2\gamma$ and $V = 10\gamma$. Physical product-state perturbations resolve an exchange of high-branch response on the monostable side of the left fold. A biorthogonal projection identifies the invariant coupling product, while coherence elimination recovers the pair as two self-consistent roots of one effective eigenvalue branch. We apply established projection and nonlinear spectral methods to identify this model-specific response exchange, root structure, and dynamical limits. We restrict claims to the reported sizes and do not infer an asymptotic switching action or a universal fold-to-EP correspondence.

II. MODEL, MEAN-FIELD FOLD, AND FINITE-SIZE SPECTRUM

We consider N two-level atoms with excited-state number $\hat{N}_e = \sum_i \hat{n}_i$, collective spin $\hat{J}_\alpha = \frac{1}{2}\sum_i \hat{\sigma}_i^\alpha$, and independent decay:

$$\dot{\rho} = \mathcal{L}_N\rho = -i[\hat{H},\rho] + \gamma\sum_i \mathcal{D}[\hat{\sigma}_i^-]\rho, \tag{1}$$

$$\hat{H} = -\Delta\hat{N}_e + \Omega\hat{J}_x + \frac{V}{2N}\hat{N}_e(\hat{N}_e - 1), \tag{2}$$

where $\mathcal{D}[c]\rho = c\rho c^\dagger - \{c^\dagger c,\rho\}/2$. This is the uniformly interacting, locally damped model of Refs. [1, 2], within

the exactly solvable steady-state family of Ref. [12]. The Supplemental Material (SM) gives the finite-N parameter mappings, including the detuning shift associated with $\hat{N}_e(\hat{N}_e-1)$. We set $\gamma=1$, so frequencies and eigenvalues below are measured in units of γ, and time in γ^{-1}.

Permutation invariance reduces the operator space to dimension $\binom{N+3}{3}$ [22]. We retain every Dicke j block and the transitions between blocks caused by local decay. This differs from the maximum-spin effective-jump approximation used in recent Rydberg switching calculations [23]. Throughout, "full generator" denotes this complete permutation-invariant operator space, which contains the product states and observables considered here. It does not enumerate modes in other permutation-symmetry sectors.

In the large-N factorized limit [1, 2], write the single-atom coherence as $\rho_{eg}=u+iv$ and the excited population as n. The mean-field equations are

$$\begin{aligned}\dot{n}&=-n-\Omega v,\\ \dot{u}&=-(\Delta-Vn)v-u/2,\\ \dot{v}&=(\Delta-Vn)u-\Omega(1-2n)/2-v/2.\end{aligned}\tag{3}$$

Their stationary populations satisfy

$$F(n)=n[(\Delta-Vn)^2+1/4+\Omega^2/2]-\Omega^2/4=0.\tag{4}$$

At $\Omega=2$, $V=10$, the left fold is at

$$(\Delta_f,n_f)=(3.5996368905,\,0.1569270403).\tag{5}$$

Using $F=\partial_n F=0$, the characteristic polynomial of the Bloch Jacobian at a fold reduces to

$$\chi(\lambda)=\lambda[(\lambda+1)^2+\Omega^2/2].\tag{6}$$

Thus the zero eigenvalue is simple, including when the fold reaches the cusp. Li, Sun, and Yi [6] obtained square-root and cube-root singularities of stationary excitation densities near fold and cusp points. Those exponents describe roots of the nonlinear stationary equation; Eq. (6) concerns the distinct linear stability problem. In particular, a triple stationary root does not imply an EP3 of this Jacobian. The coalescing fixed points and the surviving high-population branch must also be distinguished; Eq. (6) describes the former.

We locate a local minimum of the separation between two real decay eigenvalues and denote it by $\Delta_*(N)$. Define

$$g_N=\frac{\lambda_+(\Delta_*)-\lambda_-(\Delta_*)}{2}>0.\tag{7}$$

The positions move from $\Delta_*(24)=3.13740295$ to $\Delta_*(56)=3.30652148$, and the half-gap decreases from 0.0292933152 to 0.000330979777 [Fig. 1]. A least-squares slope of $\ln g_N$ over $N=24,28,\ldots,56$ gives $\sigma_{24:56}=0.14050$. Restricting to $N=40$–56 gives 0.14305, while successive four-atom slopes increase from 0.13218 to 0.14383. We therefore use "approximately exponential" only for the observed size window. The data do not determine a limiting exponent or prove $\Delta_*(N)\to\Delta_f$.

The small splitting must not be confused with a slow decay rate. At $N=32$ and $\Delta_*=3.205747775$, the real pair is

$$\lambda_-=-0.765098904,\qquad \lambda_+=-0.745092273,\tag{8}$$

whereas an oscillatory pair occurs at

$$\lambda_{\rm osc}=-0.725036287\pm 2.864794973i.\tag{9}$$

The oscillatory pair decays more slowly than both real modes. Thus g_N is an excited-mode separation, distinct from the ordinary Liouvillian gap $\Gamma_{\rm diss}=-\max_{\lambda\neq0}\mathrm{Re}\,\lambda$. It does not measure the slowest relaxation toward the stationary sector.

III. PHYSICAL RESPONSE AND MODE CHARACTER

A Hermitian right eigenoperator at a nonzero decay eigenvalue is traceless; its signed excitation-number profile is not a probability distribution. We test mode character through observable residues, whose dependence on both preparation and detection is essential in nonnormal dynamics [10, 21].

Let R_i and W_i be right and transpose-left eigenvectors of the stored generator, normalized by $W_i^T R_j=\delta_{ij}$. The transpose denotes the algebraic dual of the chosen vectorization. For the excitation-density observable $o^T\rho=\langle\hat{N}_e\rangle/N$, an initial difference $\delta\rho$ gives the exact contribution of a simple mode,

$$\delta n_i(t)=a_i e^{\lambda_i t},\qquad a_i=(o^T R_i)(W_i^T\delta\rho).\tag{10}$$

Each residue is invariant under reciprocal rescaling of its left and right eigenvectors. These signed weights are not probabilities of physical states or metastable phases.

We use two physical preparations, whose construction is specified in the SM. The first subtracts two positive product states displaced by $\pm\eta$, with $\eta=0.005$, along the real relaxation direction of the high mean-field fixed point. The second subtracts the exact stationary state from a product state prepared at the left-fold Bloch vector. This provides a controlled low-population preparation without assuming a finite-size metastable state outside the bistable interval.

For $N=32$, at $\delta=\Delta-\Delta_*=-0.06$, the high-branch perturbation has real-mode residues

$$(a_-,a_+)=(8.94\times10^{-6},\,1.1694\times10^{-3}).\tag{11}$$

At $\delta=+0.06$, they become

$$(a_-,a_+)=(1.1082\times10^{-3},\,-2.91\times10^{-6}).\tag{12}$$

The dominant high-branch response transfers from λ_+ to λ_- [Fig. 2(a)]. A separate $N=40$ calculation gives

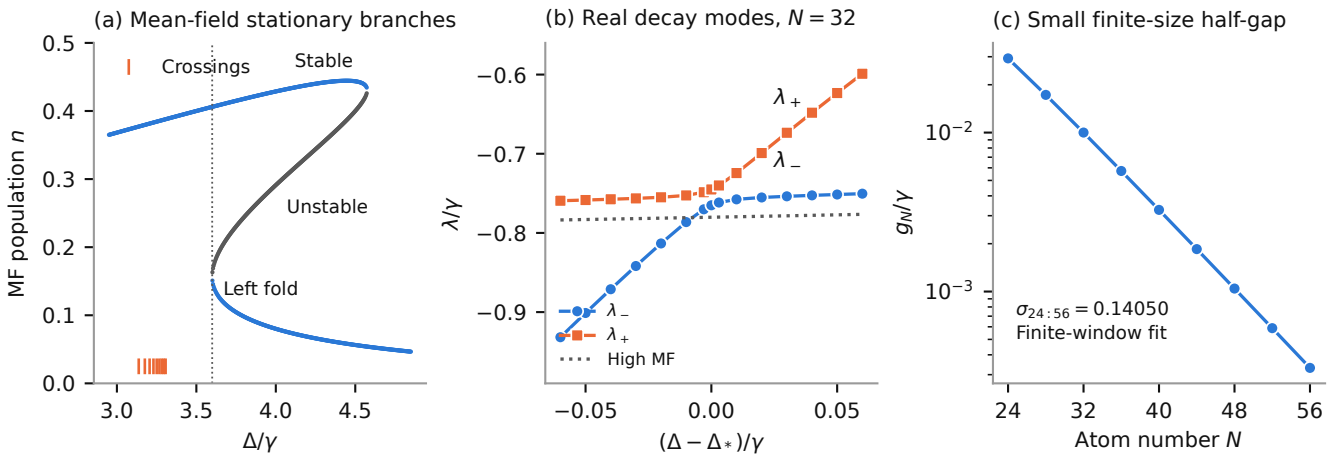


FIG. 1. Mean-field geometry and the finite-size real-mode crossing at $\Omega = 2$, $V = 10$. (a) Stationary populations: stable branches in blue, unstable branch in gray. Orange ticks mark the real-axis crossing positions for $N = 24, 28, \ldots, 56$; all lie on the monostable side of the left fold. (b) The two target real eigenvalues for $N = 32$, ordered as $\lambda_- < \lambda_+$, versus $\delta = \Delta - \Delta_*$. The dotted curve is the real relaxation eigenvalue of the surviving high mean-field branch. The modes exchange high-branch response, but the finite-size rates need not equal the mean-field rate. (c) Half-gap at its local minimum. The quoted exponent is a finite-window fit, not an established large-N limit.

the same exchange, with the nondominant residue below 7.3×10^{-7} at either endpoint. Halving η changes the normalized residue vector by less than 7.3×10^{-6} in relative Euclidean norm.

The fold-state preparation excites both modes, with large opposite residues near the crossing [Fig. 2(b)]. At $\delta = 0$, they are approximately $(-1.666, 1.423)$ for $N = 32$ and $(-5.475, 5.217)$ for $N = 40$. Opposite residues can cancel in a bounded physical signal, as analyzed for collective-spin exceptional spectral phases [21]. Our values establish neither a $1/g_N$ residue asymptote nor a real-axis Jordan limit; the SM states the additional limiting conditions.

The second diagnostic is the signed excitation-number profile $P_i(k) = \mathrm{Tr}(\Pi_k R_i)$. We measure its absolute-profile fraction in the region $k/N < 0.22$, using the same fixed diagnostic cutoff on both sides. For $N = 32$, the fractions change from $(0.209, 0.0305)$ at $\delta = -0.06$ to $(0.0314, 0.237)$ at $\delta = +0.06$ [Fig. 2(c)]. Thus the branch with greater low-density support exchanges places with the high-branch response. Neither profile is purely localized in that region. We describe the second branch as having enhanced low-density character; a rigorous identification with a universal ghost-escape mode would require an independently controlled escape problem and is not assumed here.

IV. BIORTHOGONAL TWO-MODE REDUCTION AND COMPLEX-DETUNING EPS

A. Local reduction at the gap minimum

Write $\mathcal{L}_N(\Delta) = L_0 + \Delta L_1$, with $L_1\rho = i[\hat{N}_e, \rho]$. At Δ_*, choose bases R, W of the exact two-dimensional right and left invariant subspaces with $W^T R = I$. Diagonalizing $W^T L_1 R$ defines a local diabatic basis with distinct real slopes s_a, s_b. Holding that basis fixed gives

$$H_P(\delta) = \begin{pmatrix} e_a + s_a\delta & V_{ab} \\ V_{ba} & e_b + s_b\delta \end{pmatrix}, \qquad \delta = \Delta - \Delta_*. \quad (13)$$

Here a denotes the smaller-absolute-slope branch. At $\delta = 0$ the projection is exact. Coupling to the complementary invariant subspace produces a spectral self-energy starting at order δ^2 (SM); its small absolute size does not alone establish its smallness relative to an exponentially small coupling.

The usual two-by-two discriminant [24] gives the squared separation and its derivative at the projection point:

$$d^2 = (e_a - e_b)^2 + 4V_{ab}V_{ba}, \qquad \partial_\Delta d^2 = 2(e_a - e_b)(s_a - s_b). \quad (14)$$

At a nondegenerate minimum of a positive real gap, distinct slopes imply $e_a = e_b$ and hence

$$\boxed{g_N^2 = V_{ab}V_{ba}.} \quad (15)$$

This identity uses the exact finite-size spectral subspace. It organizes the gap but is not an independent prediction of its exponent.

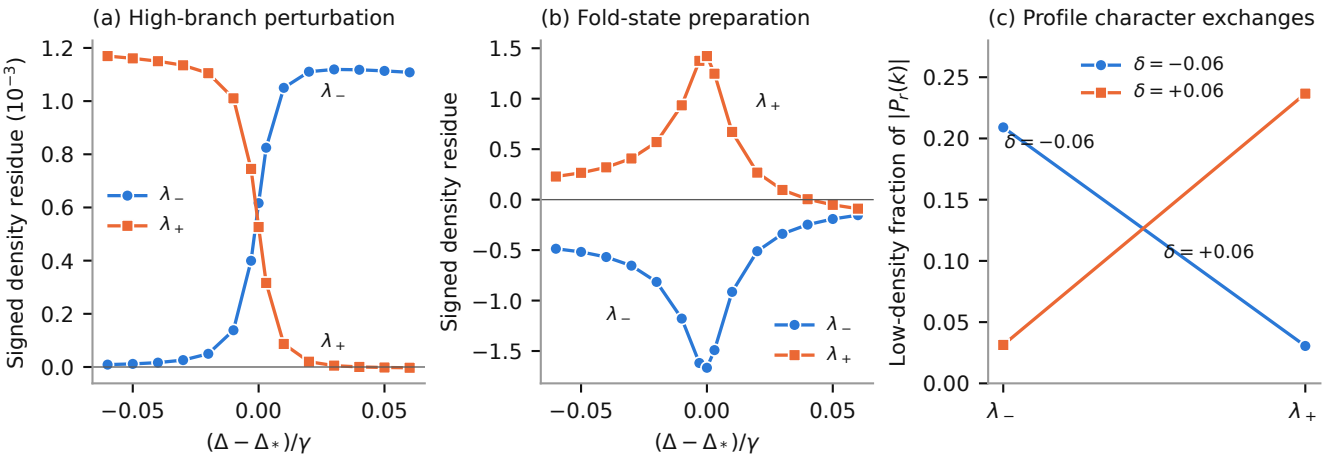


FIG. 2. Physical residues and exchange of mode character, $N = 32$. (a) Residues of the two real modes in the excitation-density difference between product states displaced on opposite sides of the high mean-field fixed point. (b) Residues for a product state prepared at the left-fold Bloch vector, relative to the finite-size stationary state. Large opposite residues near the crossing cancel in the physical signal. (c) Fraction of the absolute excitation-number profile below $k/N = 0.22$ at the two endpoints. This ratio is independent of mode normalization and characterizes the signed profile; it is not a population probability of a physical state.

Under $R_i \mapsto c_i R_i$, $W_i \mapsto c_i^{-1} W_i$,

$$V_{ab} \mapsto (c_b/c_a)V_{ab}, \qquad V_{ba} \mapsto (c_a/c_b)V_{ba}. \tag{16}$$

Individual coupling exponents can therefore depend on normalization if c_i depend on N. We fix a population-peak convention, $\max_k |P_i(k)| = 1$, with the largest-magnitude entry positive. Over $N = 24$–56, the descriptive exponents are

$$\alpha_{ab} = 0.009638, \qquad \alpha_{ba} = 0.271365,$$
$$(\alpha_{ab} + \alpha_{ba})/2 = 0.140501. \tag{17}$$

Normalizing the right operators by their physical trace norm changes the first two values to 0.003563 and 0.277440, while preserving their sum. A balanced convention makes both equal to σ by construction [Fig. 3]. Thus a directional exponent cannot be assigned an invariant activation interpretation from this decomposition alone.

The weak matrix element contains cancellations involving coherences. For $N = 40$, summing its output-diagonal and output-off-diagonal contributions separately gives

$$V_{ba} = 0.9659542705 - 0.9657905699 = 1.63700625 \times 10^{-4}. \tag{18}$$

This is not a representation-independent separation into two physical processes. Indeed, $W_b^T(L - cI)R_a = W_b^T L R_a$ for any scalar c, and choosing $c = e_a$ moves the cancellation into the small residual $(L - e_a)R_a$. The robust conclusion is that a right population tail by itself does not determine the full biorthogonal matrix element. A boundary-flux identity and its dependence on the chosen cut are given in the SM.

B. Analytic continuation in detuning

We separately continue the full generator as $L_0 + \zeta L_1$ for complex detuning ζ. This continuation uses $-i[H(\zeta), \rho]$; it is not $-i[H\rho - \rho H^\dagger]$. It remains trace preserving but is generally not Hermiticity preserving or CPTP.

A local two-subspace Newton calculation gives the upper-half-plane EP positions

$$\zeta_{\mathrm{EP}}(24) = 3.136806063 + 0.026370766i,$$
$$\zeta_{\mathrm{EP}}(32) = 3.205686613 + 0.007464558i, \tag{19}$$
$$\zeta_{\mathrm{EP}}(40) = 3.250218904 + 0.002129005i.$$

At each point, the projected traceless matrix M has nonzero norm, while $\|M^2\|_F / \|M\|_F^2 < 3.3 \times 10^{-15}$, consistent with a rank-one nilpotent. For $N = 32$, analytic continuation around a circle of radius 0.00223937 exchanges the two full-generator eigenvalues after one circuit and restores them after two. Changing the eigensolver seed and shift and doubling the sampling density preserves the result (SM).

The local linear model predicts $|\operatorname{Im}\zeta_{\mathrm{EP}}| \simeq 2g_N/|s_a - s_b|$. The relative differences in the imaginary parts are 0.456%, 0.0792%, and 0.0115% for $N = 24, 32, 40$, respectively. This verifies nearby complex partners for the studied crossings; we do not claim that an arbitrary avoided crossing necessarily has a nearby EP. Nor does an EP in complex detuning constitute a physical degeneracy at real parameters.

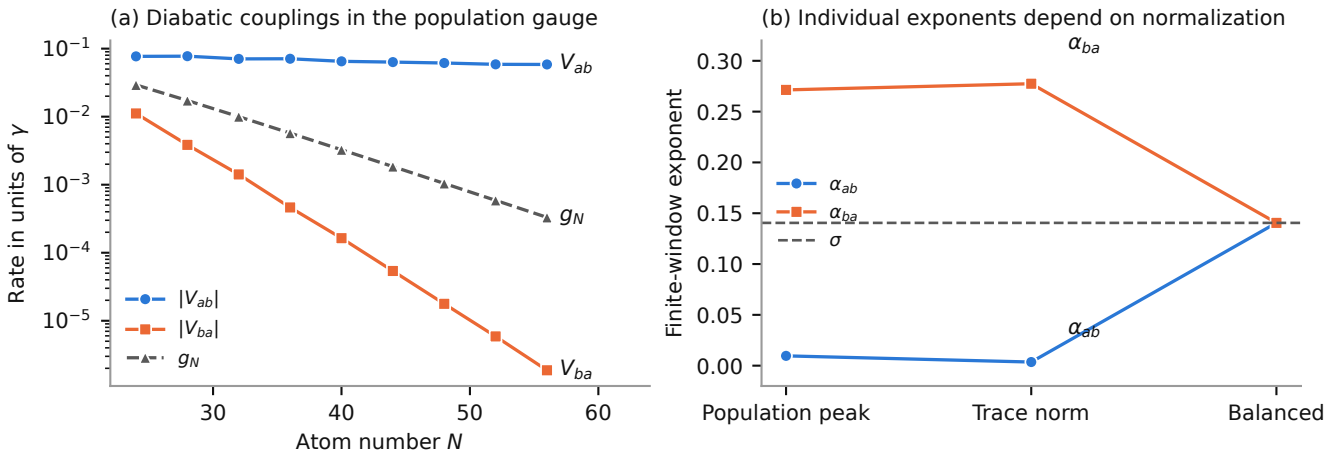


FIG. 3. Biorthogonal gap decomposition. (a) Couplings in the population-peak convention, together with the invariant half-gap. (b) Finite-window exponents under three normalizations of the same two-dimensional subspace. The physical trace-norm convention changes the individual exponents; balancing the couplings makes each equal to σ by construction. The invariant statement is their product and the sum of their logarithmic slopes.

V. EXACT ELIMINATION OF COHERENCE SECTORS

A. Spectral-parameter-dependent effective operator

We use projection methods that retain the spectral parameter [25, 26], evaluated by matrix continued fractions as in spin master equations [27]. Group the Dicke operator basis by signed coherence order $q = m - m'$. The diagonal Hamiltonian and local decay preserve q, whereas the drive connects q only to $q \pm 1$. This gives a block-tridiagonal generator while retaining every Dicke j block.

Let $A_q = L_{q,q}$, $U_q = L_{q,q+1}$, and $D_q = L_{q+1,q}$. Starting at the highest positive order $q_{\mathrm{top}} = N$, define

$$\Sigma_{q_{\mathrm{top}}}(z) = 0,$$
$$\Sigma_{q-1}(z) = U_{q-1}[zI - A_q - \Sigma_q(z)]^{-1} D_{q-1}. \tag{20}$$

Here z is an eigenvalue variable, distinct from the complex detuning ζ above. For real physical parameters and real z, Hermitian conjugation relates the positive and negative coherence chains. Their sum yields

$$K_N(z) = A_0 + 2\,\mathrm{Re}\,\Sigma_0(z), \qquad K_N(z) r_0 = z r_0. \tag{21}$$

The $q = 0$ sector still contains the populations of every Dicke block; it is not the excitation-number marginal alone, and $K_N(z)$ need not be a classical Markov generator.

With P projecting onto $q = 0$ and $Q = I - P$, the Schur identity reads

$$\det(zI - L) = \det(zI - L_{QQ})\det[zI - K_N(z)]. \tag{22}$$

It applies where $z - L_{QQ}$ is invertible; the recursion also requires invertible intermediate denominators. Purely coherent modes with zero $q = 0$ projection and points at elimination poles require separate treatment. For complex z, the analytic continuation involves $\Sigma_+(z) + \overline{\Sigma_+(\bar{z})}$, not $2\,\mathrm{Re}\,\Sigma_+(z)$. Our real-axis recursion is not used to compute Eq. (19).

The spectral dependence has a time-domain counterpart in the projection-operator memory formalism [28, 29]. Direct elimination of x_Q from our linear equations gives

$$\dot{x}_0(t) = L_{00} x_0(t) + L_{0Q} e^{t L_{QQ}} x_Q(0) + \int_0^t L_{0Q} e^{(t-s) L_{QQ}} L_{Q0} x_0(s)\, ds. \tag{23}$$

Thus the self-energy represents memory carried by the eliminated coherence sectors. Replacing it by a constant is an additional approximation.

B. Two roots of one effective branch

In the local real spectral window, select a simple real eigenvalue branch $\mu_N(z)$ of $K_N(z)$ and define $f_N(z) = \mu_N(z) - z$. We find a negative local minimum,

$$f_N(z_c) = -\epsilon_N < 0, \qquad f_N'(z_c) = 0,$$
$$c_N = f_N''(z_c) > 0. \tag{24}$$

Its two nearby roots give the two decay eigenvalues. Expanding around the minimum gives

$$f_N(z) = -\epsilon_N + \frac{c_N}{2}(z - z_c)^2 + \cdots, \qquad g_{N,\mathrm{quad}} = \sqrt{\frac{2\epsilon_N}{c_N}}. \tag{25}$$

The derivatives are calculated analytically through the matrix recursion and the non-Hermitian effective eigenproblem (SM).

TABLE I. Local real-axis coherence reduction. g_N is the independently computed full-generator half-gap; the last column is $g_{N,\mathrm{quad}}/g_N - 1$. All entries use $\gamma = 1$.

N	ϵ_N	c_N	g_N	Relative error
24	9.33404×10^{-3}	22.28695	0.02929332	-1.2002×10^{-2}
32	1.09619×10^{-3}	21.96534	0.01000332	-1.2773×10^{-3}
40	1.28422×10^{-4}	24.11955	0.00326360	-1.0842×10^{-4}
48	1.51689×10^{-5}	27.82335	0.00104421	-6.5891×10^{-6}

For $N = 24, 32, 40, 48$, direct roots of $f_N = 0$ reproduce the independent full-generator half-gaps to relative differences below 1.3×10^{-9}. Back-substituting all eight roots into the full generator gives $\|LR - zR\|_2/\|R\|_2 < 1.2 \times 10^{-13}$. Table I and Fig. 4 show the separate, larger error of the quadratic approximation. The minimum location is not generally the midpoint of the two roots: at $N = 24$, these are $z_c = -0.745502759$ and $(z_- + z_+)/2 = -0.750910792$.

The two roots do not correspond to two nearby eigenvalues of a frozen $K_N(z_c)$. For $N = 48$, its nearest eigenvalues include -0.75910208, -0.21846755, and zero, whereas the full generator has the two roots -0.76013902 and -0.75805059. The root pair is retained by self-consistency in z. This finding concerns this specific elimination; it does not rule out a different, correctly matched finite-dimensional effective model.

A four-point fit of $\ln \epsilon_N$ gives $\alpha_\epsilon = 0.267635$, while the same convention applied to the curvature gives $\alpha_c = -0.009490$. Consequently the quadratic gap has fitted exponent $(\alpha_\epsilon - \alpha_c)/2 = 0.138562$, compared with 0.139030 for the exact gap over the same four sizes. These are different fit windows from $\sigma_{24:56}$ above. No argument yet fixes the large-N scaling of ϵ_N or proves that c_N contributes only an algebraic prefactor. Equation (25) is therefore a tested finite-size description, not an action-based exponent prediction.

C. Connection to biorthogonal self-orthogonality

The effective spectral slope has a precise relation to the full-mode norm. Write $G = (zI - L_{QQ})^{-1}$, $B = L_{0Q}$, and $C = L_{Q0}$, so that $K = L_{00} + BGC$ and $K_z = -BG^2C$. For a simple effective eigenbranch with $Kr = \mu r$, $l^T K = \mu l^T$, and $l^T r = 1$, define the back-substituted vectors

$$R = \begin{pmatrix} r \\ GCr \end{pmatrix}, \qquad W^T = (l^T,\, l^T BG). \tag{26}$$

Then

$$\boxed{W^T R = l^T (I - K_z) r = 1 - \mu_z = -f_z.} \tag{27}$$

At a root, this is the standard derivative pairing for the nonlinear eigenvalue problem $T(z) = zI - K(z)$ [26, 30]. Its appearance here follows from Schur back-substitution, rather than a new normalization principle. Away from a root, the vectors instead satisfy

$$(L - zI)R = \begin{pmatrix} f(z)r \\ 0 \end{pmatrix}, \qquad W^T(L - zI) = (f(z)l^T, 0). \tag{28}$$

Thus $f_z(z_c) = 0$ at the observed negative minimum does not make z_c an EP: $f(z_c) \neq 0$, so it is not an eigenvalue.

For comparison, if the elimination is regular, the selected eigenvalue of $K(z_c)$ is algebraically simple, and $f = f_z = 0$ with $f_{zz} \neq 0$, Eq. (22) implies an exactly double characteristic root. The kernel remains one-dimensional under back-substitution, giving an EP2. This conditional statement uses the analytic Schur operator and connects nonlinear spectral self-consistency to the usual defective-eigenvalue criterion. It does not supply the small depth ϵ_N in the present real-axis problem.

VI. DYNAMICAL REACH AND LIMITS OF THE REAL-MODE PAIR

Spectral hybridization does not ensure that two modes describe a measured transient. Large expansion coefficients and preparation-dependent visibility make a direct dynamical comparison necessary [10, 11, 21]. We sum Eq. (10) over the two real modes and compare with $e^{tL}\delta\rho$ for both preparations. At $N = 32$, we also include the oscillatory conjugate pair in Eq. (9). Sparse exponential multiplication provides the reference evolution, without fitting transient amplitudes or rates.

At the crossing, the relative L^2 error over $6 \le t \le 12$ for the high-branch preparation is 25.1% with the two real modes and 0.506% with all four modes. For the fold-state preparation the corresponding errors are 69.5% and 5.00%. The latter initial condition has larger contributions from other modes and stronger cancellations; the same four-mode truncation need not be equally accurate for different preparations. On the two sides $\delta = \pm 0.06$, the high-branch four-mode errors over the same window remain below 0.53%.

Figure 5 removes the common real-pair decay envelope only for visibility. Error definitions are in the SM, and the unscaled signals are supplied with the source data. The oscillatory pair is needed to account for the observed oscillations over the tested time window. The real-pair mean decay rate remains of order γ, so its small splitting alone establishes no long-lived response. Possible size-dependent transient enhancement would also require analyzing residues; it cannot be decided from these eigenvalues alone. The present tests support the specific response exchange and quantify the two-mode truncation's error.

VII. DISCUSSION AND CONCLUSIONS

We have resolved a finite-size exchange of physical relaxation response in a Rydberg model retaining inde-

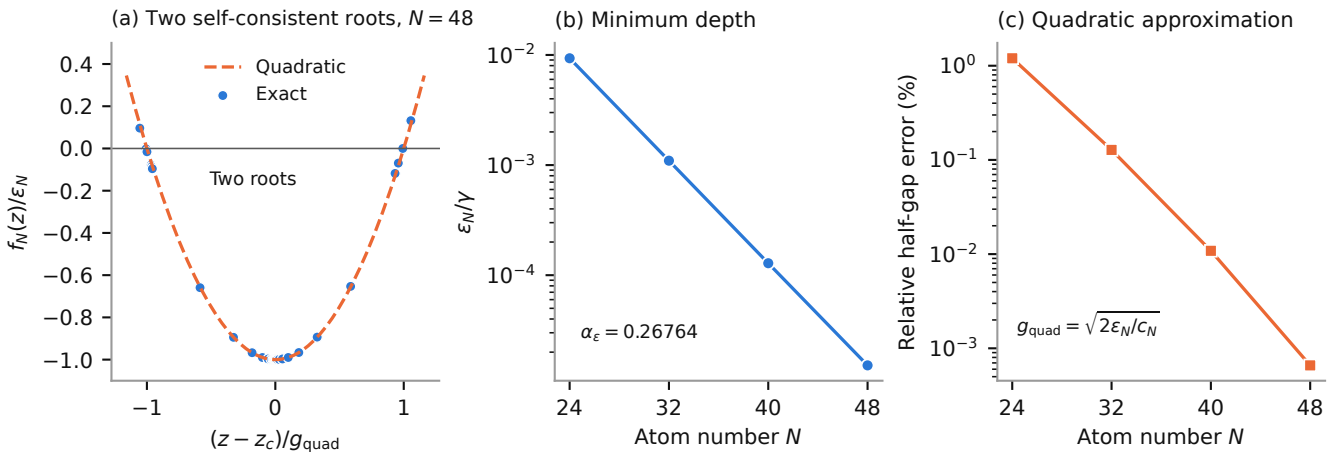


FIG. 4. Exact coherence elimination at the supplied real-axis crossing detunings. (a) The normalized self-consistency function for $N = 48$ (points) and its local quadratic expansion (dashed). (b) Negative-minimum depth and its descriptive four-size exponent. (c) Relative error of the quadratic half-gap. The exact eliminated roots have much smaller errors than this local expansion. Both the depth and curvature are outputs of a finite-N spectral calculation; no independent large-N action is inferred.

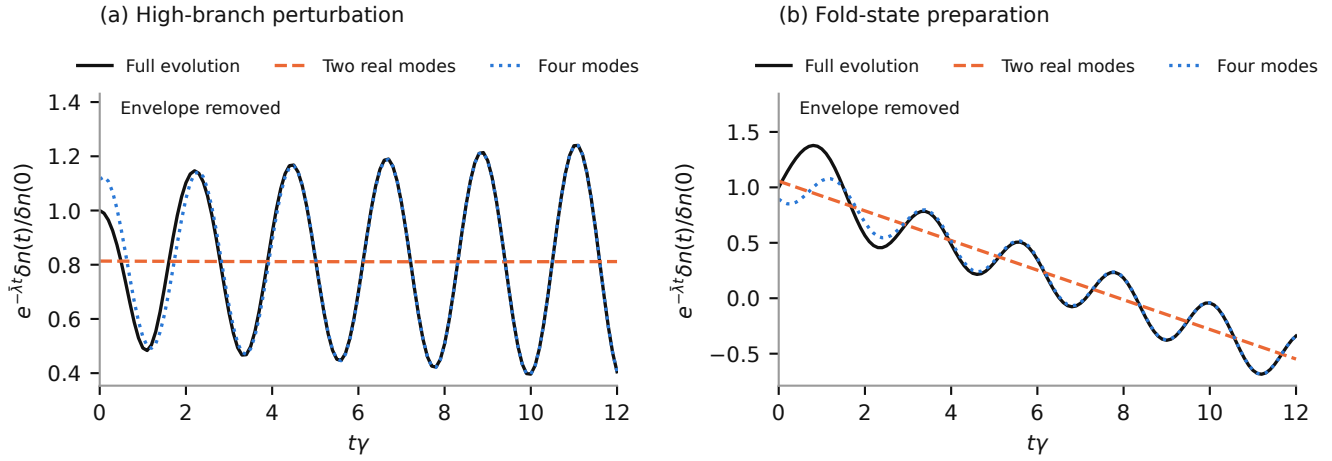


FIG. 5. Physical density transients at the $N = 32$ crossing. Solid black: full permutation-invariant evolution. Orange dashed: the two target real modes with exact spectral residues. Blue dotted: those modes plus the oscillatory conjugate pair. (a) High-branch product-state perturbation; (b) fold-state preparation relative to the stationary state. The plotted signal is divided by its initial value and multiplied by $e^{-\bar{\lambda}t}$, with $\bar{\lambda} = (\lambda_- + \lambda_+)/2$, to expose spectral contributions after removing the common decay envelope. Errors quoted in the text are calculated from the unscaled signals.

pendent local decay and all permutation-invariant spin blocks. Prior collective-spin studies establish finite-rate doublets and cancelling residues [20, 21]; here physical preparations identify a detuning-dependent exchange near the mean-field fold. The invariant coupling product and two self-consistent Schur roots quantify its splitting using established methods. Freezing the Schur operator loses the pair, while a slower oscillatory pair is required for the tested transients. Together, these results separate spectral hybridization from a universal two-mode relaxation law.

The unresolved asymptotic target is the scaling of ϵ_N/c_N, or equivalently $V_{ab}V_{ba}$, independently of the exact pair. Exact steady-state information is available [12], and Noachtar and Clerk [16] conjecture a barrier relation using a purification selected by hidden time-reversal symmetry. That conjecture concerns the ordinary Liouvillian gap, not our finite-rate splitting or an arbitrary excitation-number marginal potential $\Phi(n_f)$. Likewise, applying switching-instanton theory [14, 15] to this excited pair requires a controlled construction for that spectral observable. Numerical agreement between finite-window exponents supplies neither derivation.

Our conclusions concern the reported finite sizes, uni-

form interactions, and local decay. The data do not establish a thermodynamic fold correspondence, a universal ghost-escape mode, or short-range switching. Size-dependent transient enhancement also remains a separate question requiring residues, rather than eigenvalues alone [10, 11].

DATA AND NUMERICAL METHODS

The SM specifies the generator, preparations, projection identities, and numerical checks. Plotted records and reproduction scripts accompany the source bundle. Independent full-spin checks cover $N = 1$–4; the finite-size calculations use no fitted stochastic noise model.

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